Parametric Equations & Modeling

ANSWER KEY

Section 1: Evaluating Parametric Equations and Graph Points

1 A particle moves according to $x(t) = t^2 - 3$ and $y(t) = 2t + 1$. What is its position at $t = 3$?

a. $(6, 7)$ ✓

b. $(9, 7)$

c. $(6, 6)$

d. $(3, 7)$

$x(3) = 3^2 - 3 = 6$. $y(3) = 2(3) + 1 = 7$. The position point is $(6, 7)$.

2 At what value of t does the parametric curve $x(t) = t^2 - 4$, $y(t) = t - 1$ intersect the y-axis with $y > 0$?

a. $t = 2$ ✓

b. $t = -2$

c. $t = 4$

d. $t = 1$

The curve crosses the y-axis when $x(t) = 0 \Rightarrow t^2 - 4 = 0 \Rightarrow t = \pm 2$. For $t = 2$, $y = 2 - 1 = 1 > 0$. For $t = -2$, $y = -3 < 0$. Thus $t = 2$.

3 Given $x(t) = 4\cos t$ and $y(t) = 4\sin t$ for $0 \le t \le 2\pi$, what curve is traced and in which direction?

a. **A circle of radius 4 traced counterclockwise** ✓

b. A circle of radius 4 traced clockwise

c. An ellipse of semi-major axis 4

d. A line segment from $(-4, -4)$ to $(4, 4)$

At $t = 0$, position is $(4, 0)$. At $t = \frac{\pi}{2}$, position is $(0, 4)$. As t increases, motion proceeds counterclockwise around a circle of radius 4.

Section 2: Eliminating the Parameter

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4 Eliminate the parameter t from $x(t) = 3t + 2$ and $y(t) = 6t - 5$.

a. $y = 2x - 9$ ✓

b. $y = 2x - 7$

c. $y = 2x + 1$

d. $y = 3x - 11$

Solve for t from x: $t = \frac{x-2}{3}$. Substitute into y: $y = 6\left(\frac{x-2}{3}\right) - 5 = 2(x-2) - 5 = 2x - 4 - 5 = 2x - 9$.

5 Eliminate the parameter from $x = 3\cos\theta$ and $y = 5\sin\theta$.

a. $\frac{x^2}{9} + \frac{y^2}{25} = 1$ ✓

b. $\frac{x^2}{25} + \frac{y^2}{9} = 1$

c. $x^2 + y^2 = 34$

d. $9x^2 + 25y^2 = 1$

Isolate trig functions: $\cos\theta = \frac{x}{3}$ and $\sin\theta = \frac{y}{5}$. Using $\cos^2\theta + \sin^2\theta = 1$ yields $\frac{x^2}{9} + \frac{y^2}{25} = 1$.

6 Eliminate the parameter from $x = t^2$ and $y = t^4 + 3$ for $t \in (-\infty, \infty)$. What is the resulting Cartesian equation and its domain?

a. $y = x^2 + 3$ **with domain** $x \geq 0$ ✓

b. $y = x^2 + 3$ with domain $(-\infty, \infty)$

c. $y = 2x + 3$ with domain $x \geq 0$

d. $y = x^4 + 3$ with domain $x \geq 0$

Since $y = (t^2)^2 + 3 = x^2 + 3$. Because $x = t^2$, x cannot be negative; the domain is restricted to $x \geq 0$.

Section 3: Projectile Motion Modeling

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7 A projectile is launched with initial speed $v_0 = 80$ ft/s at an angle of 30° from the horizontal from ground level. Which parametric equations model its path ($g = 32$ ft/s^2)?

a. $x(t) = 40\sqrt{3}\,t$ **and** $y(t) = 40t - 16t^2$ ✓

b. $x(t) = 40t$ and $y(t) = 40\sqrt{3}\,t - 16t^2$

c. $x(t) = 80t$ and $y(t) = 80t - 32t^2$

d. $x(t) = 40\sqrt{3}\,t$ and $y(t) = 40t - 32t^2$

$v_{0x} = 80\cos(30^\circ) = 80\left(\frac{\sqrt{3}}{2}\right) = 40\sqrt{3}$. $v_{0y} = 80\sin(30^\circ) = 80\left(\frac{1}{2}\right) = 40$. In feet, $y(t) = v_{0y}t - \frac{1}{2}gt^2 = 40t - 16t^2$.

8 For the projectile above with $y(t) = 40t - 16t^2$, what is its total flight time before landing?

a. 2.5 seconds ✓

b. 1.25 seconds

c. 5.0 seconds

d. 3.0 seconds

Set $y(t) = 0 \Rightarrow t(40 - 16t) = 0$. Discarding $t = 0$, $16t = 40 \Rightarrow t = \frac{40}{16} = 2.5$ seconds.

9 What is the maximum height attained by a projectile modeled by $y(t) = 64t - 16t^2$?

a. 64 ft ✓

b. 32 ft

c. 128 ft

d. 48 ft

Maximum height occurs at vertex $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ s. Maximum height: $y(2) = 64(2) - 16(4) = 128 - 64 = 64$ ft.

Section 4: Direction of Motion and Speed

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10 As parameter t increases from 0 to π, how does the curve $x = \sin t, y = \cos t$ move?

a. **Clockwise from** $(0, 1)$ **to** $(0, -1)$ ✓

b. Counterclockwise from $(1, 0)$ to $(-1, 0)$

c. Clockwise from $(1, 0)$ to $(-1, 0)$

d. Counterclockwise from $(0, 1)$ to $(0, -1)$

At $t = 0$: $(x, y) = (0, 1)$. At $t = \pi/2$: $(1, 0)$. At $t = \pi$: $(0, -1)$. Moving from $(0, 1)$ through $(1, 0)$ to $(0, -1)$ is clockwise.

11 If a curve is given by $x(t) = t + 1$ and $y(t) = t^2$, what is the average rate of change $\frac{\Delta y}{\Delta x}$ from $t = 1$ to $t = 3$?

a. 4 ✓

b. 8

c. 2

d. 6

At $t = 1$: $(2, 1)$. At $t = 3$: $(4, 9)$. $\frac{\Delta y}{\Delta x} = \frac{9-1}{4-2} = \frac{8}{2} = 4$.

12 Which parametric curve represents a straight line segment with endpoints $(1, 2)$ and $(7, 14)$ for $0 \le t \le 1$?

a. $x(t) = 1 + 6t, y(t) = 2 + 12t$ ✓

b. $x(t) = 7t, y(t) = 14t$

c. $x(t) = 1 + 7t, y(t) = 2 + 14t$

d. $x(t) = 6t, y(t) = 12t$

Using linear interpolation $(1 - t)P_0 + tP_1$: $x(t) = 1 + t(7 - 1) = 1 + 6t$ and $y(t) = 2 + t(14 - 2) = 2 + 12t$ for $0 \le t \le 1$.

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